ASSIGNMENT-3
Differential Calculus
Unit III

Name: _____ **Roll No.:** _____

Class: _____ **Date:** _____

Instructions:

- Attempt all questions.
- Show all necessary steps of calculation.
- Give proper justification for conceptual questions.
- Draw neat and labelled diagrams wherever required.

Part A:

Q1. Define a function of two variables. Find the domain and range of

$$f(x, y) = \sqrt{9 - x^2 - y^2}.$$

Q2. Evaluate

$$\lim_{(x,y)\to(0,0)} \frac{x^2 + y^2}{x^2 + 2y^2}.$$

Does the limit exist? Justify your answer.

Q3. Determine whether the function

$$f(x, y) = \begin{cases} \dfrac{x^2 y}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0) \end{cases}$$

is continuous at $(0, 0)$.

Q4. If

$$z = x^3 y^2 + 2x^2 y + y^3,$$

find

$$\frac{\partial z}{\partial x} \quad \text{and} \quad \frac{\partial z}{\partial y}.$$

Q5. Explain the geometrical meaning of

$$\frac{\partial z}{\partial x} \quad \text{and} \quad \frac{\partial z}{\partial y}$$

for a surface

$$z = f(x, y).$$

Give a suitable sketch or explanation.

Q6. Determine whether

$$f(x, y) = x^3 + 3x^2y + 3xy^2 + y^3$$

is a homogeneous function.

If yes, find its degree of homogeneity.

Q7. Verify Euler's theorem for the homogeneous function

$$u = x^2 + 2xy + y^2.$$

Q8. Find the total differential dz for

$$z = x^2y + xy^2.$$

Part B:

Q9. Determine whether the function

$$f(x, y) = x^2 + y^2$$

is differentiable at $(1, 2)$.

Find its total differential at that point.

Q10. If

$$z = x^2 + y^2, \qquad x = u + v, \qquad y = u - v,$$

find

$$\frac{\partial z}{\partial u} \quad \text{and} \quad \frac{\partial z}{\partial v}$$

using the chain rule.

Q11. If

$$z = e^{xy}, \qquad x = r\cos\theta, \qquad y = r\sin\theta,$$

find

$$\frac{\partial z}{\partial r} \quad \text{and} \quad \frac{\partial z}{\partial \theta}.$$

Q12. Find the Jacobian

$$\frac{\partial(u, v)}{\partial(x, y)}$$

if

$$u = x + y, \qquad v = x - y.$$

Q13. If

$$u = x^2 - y^2, \qquad v = 2xy,$$

find

$$\frac{\partial(u, v)}{\partial(x, y)}.$$

Q14. Given

$$u = x + y, \qquad v = x - y,$$

find

$$\frac{\partial(x, y)}{\partial(u, v)}$$

and verify that

$$\frac{\partial(u, v)}{\partial(x, y)} \frac{\partial(x, y)}{\partial(u, v)} = 1.$$

Q15. Using the ε-δ definition of the limit, verify that

$$\lim_{(x,y)\to(1,2)} (3x + 2y) = 7.$$

Find a suitable δ in terms of ε and prove the result directly from the ε-δ definition.

Q16. Consider the function

$$f(x, y) = |x| + |y|.$$

Determine whether f is continuous at $(0, 0)$. Also investigate whether the partial derivatives

$$f_x(0, 0) \qquad \text{and} \qquad f_y(0, 0)$$

exist at $(0, 0)$.

Hence, show that continuity of a function at a point does not necessarily imply the existence of its partial derivatives at that point.

Q17. If $u = f(x, y)$ is a homogeneous function of degree n, state Euler's theorem.

Then prove the extended result

$$x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = n(n - 1)u.$$

Q18. Expand

$$f(x, y) = e^{x+y}$$

about the point $(0, 0)$ up to terms of degree 2 using Taylor's theorem.

Q19. Obtain the Maclaurin expansion of

$$f(x, y) = \ln(1 + x + y)$$

up to terms of degree 3.

Q20. Find the Taylor expansion of

$$f(x, y) = x^2 y + xy^2$$

about the point $(1, 1)$ up to second-degree terms.

Part C:

Q19. Find the stationary points of

$$f(x, y) = x^2 + y^2 - 4x - 6y + 13.$$

Determine whether the stationary point is a maximum, minimum, or saddle point.

Q20. Find and classify all stationary points of

$$f(x, y) = x^3 + y^3 - 3xy.$$

Q21. For a function $f(x, y)$, explain the second derivative test for determining whether a stationary point is a:

(a) local maximum,

(b) local minimum,

(c) saddle point.

Use

$$D = f_{xx} f_{yy} - (f_{xy})^2$$

in your explanation.

Q22. Using the method of undetermined multipliers, find the maximum and minimum values of

$$f(x, y) = x + y$$

subject to the constraint

$$x^2 + y^2 = 1.$$

Q23. Using the method of undetermined multipliers, find the maximum and minimum values of

$$f(x, y) = xy$$

subject to

$$x^2 + y^2 = 4.$$

Q24. Trace the curve

$$y^2 = x^2(x + 1).$$

Discuss the following:

(a) symmetry,

(b) intercepts,

(c) nature of the curve near the origin,

(d) asymptotic behaviour,

(e) important points,

(f) rough sketch of the curve.

Q25. Trace the curve

$$y^2 = \frac{x^2(1-x)}{1+x}.$$

Study the following:

(a) symmetry,

(b) intercepts,

(c) region in which the curve exists,

(d) tangent at the origin,

(e) asymptotes,

(f) maximum/minimum values, if any,

(g) nature of the curve,

(h) rough sketch.

— End of Assignment —